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Chapter 19

Representing 3-D in 2-D

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Exercise 19

Question 1(i)

Number of faces in a tetrahedron is :

- 1. 3
- 2. 4
- 3. 5
- 4. 6



Chapter 19
Representing 3-D in 2-D

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Answer

Number of faces in a tetrahedron is 4.

Hence, option 2 is the correct option.

Question 1(ii)

In an octahedron; number of faces is :

1. half of the number of edges
2. 8
3. 12
4. more than the number of edges

Answer

Number of faces in an octahedron is 8.

Hence, option 2 is the correct option.

Question 1(iii)

A rectangular pyramid has :

1. 4 vertices and 4 faces
2. 6 vertices and 6 faces
3. 7 vertices and 4 faces
4. 5 vertices and 5 faces

Answer

A rectangular pyramid has 5 vertices and 5 faces.

Hence, option 4 is the correct option.

Question 1(iv)

If V stands for number of vertices, F stands for number of faces and E stands for number of edges, then :

1. $V + F + E = 2$

2. $F + E - V - 2 = 0$

3. $F + V - E = 2$

4. $F - E + V + 2 = 0$

Answer

According to Euler's formula,

$$F + V - E = 2$$

Hence, option 3 is the correct option.

Question 1(v)

If in a polyhedron, number of faces = 12 and number of vertices is 20, then the number of edges is :

1. 10

2. 20

3. 30

4. 40

Answer

According to Euler's formula,

$$F + V - E = 2$$

Given: $F = 12$

$$V = 20$$

So,

$$\Rightarrow 12 + 20 - E = 2$$

$$\Rightarrow 32 - E = 2$$

$$\Rightarrow E = 32 - 2$$

$$\Rightarrow E = 30$$

The number of edges = 30.

Hence, option 3 is the correct option.

Question 2

If a polyhedron, has 10 vertices and 7 faces, find the number of edges in it.

Answer

Given:

Vertices, $V = 10$

Faces, $F = 7$

According to Euler's formula,

$$F + V - E = 2$$

So,

$$\Rightarrow 7 + 10 - E = 2$$

$$\Rightarrow 17 - E = 2$$

$$\Rightarrow E = 17 - 2$$

$$\Rightarrow E = 15$$

Hence, number of edges = 15.

Question 3

State the number of faces, number of vertices and the number of edges of :

(i) a pentagonal pyramid

(ii) a hexagonal prism

Answer

(i) A pentagonal pyramid

Number of faces = 6

Number of vertices = 6

Number of edges = 10

(ii) A hexagonal prism

Number of faces = 8

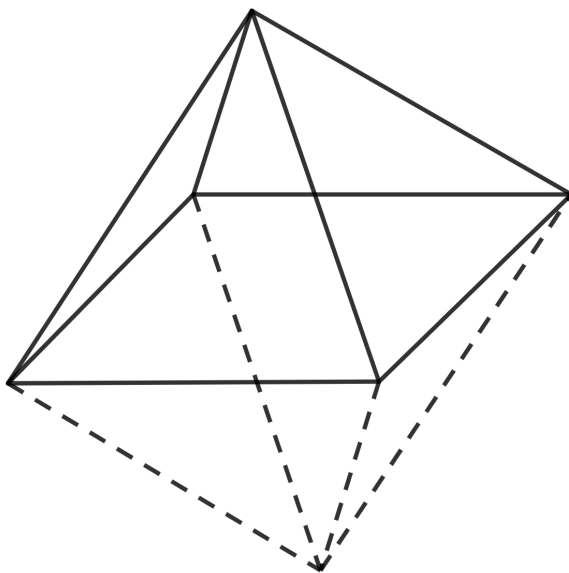
Number of vertices = 12

Number of edges = 18

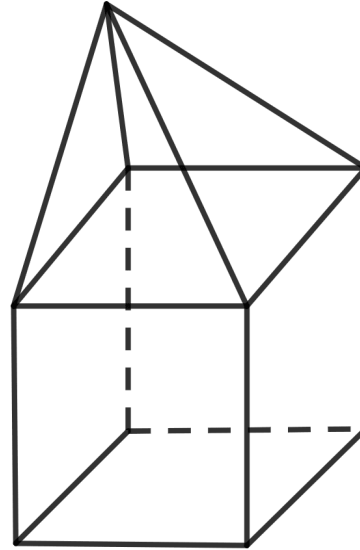
Question 4

Verify Euler's formula for the following three dimensional figures :

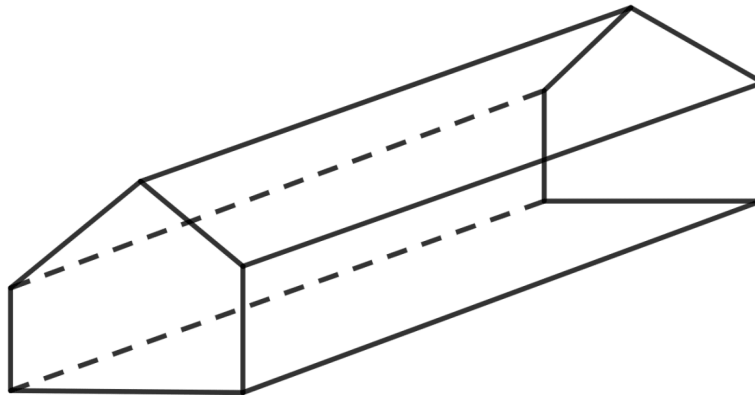
(i)



(ii)



(iii)



Answer

(i)

Given:

$$F = 8, V = 6, E = 12$$

According to Euler's formula,

$$F + V - E = 2$$

So, LHS:

$$\Rightarrow 8 + 6 - 12$$

$$\Rightarrow 14 - 12$$

$$\Rightarrow 2$$

Thus, LHS = RHS

Hence, Euler's formula is verified.

(ii)

Given:

$$F = 9, V = 9, E = 16$$

According to Euler's formula,

$$F + V - E = 2$$

So, LHS:

$$\Rightarrow 9 + 9 - 16$$

$$\Rightarrow 18 - 16$$

$$\Rightarrow 2$$

Thus, LHS = RHS

Hence, Euler's formula is verified.

(iii)

Given:

$$F = 7, V = 10, E = 15$$

According to Euler's formula,

$$F + V - E = 2$$

So, LHS:

$$\Rightarrow 7 + 10 - 15$$

$$\Rightarrow 17 - 15$$

$$\Rightarrow 2$$

Thus, LHS = RHS

Hence, Euler's formula is verified.

Question 5

Can a polyhedron have 8 faces, 26 edges and 16 vertices ?

Answer

Clearly, $F = 8$

$$V = 16$$

$$E = 26$$

According to Euler's formula,

$$F + V - E = 2$$

$$F + V - E = 8 + 16 - 26$$

$$= 24 - 26$$

$$= -2$$

$\Rightarrow F + V - E$ is not equal to 2.

Hence, a polyhedron can not have 8 faces, 26 edges and 16 vertices.

Question 6

Can a polyhedron have :

- (i) 3 triangles only ?
- (ii) 4 triangles only ?
- (iii) a square and four triangles?

Answer

(i) No, a polyhedron cannot have only 3 triangles as its faces because a polyhedron must have at least four faces.

(ii) Yes, a polyhedron can have 4 triangles, which is known as a pyramid with a triangular base, because all the edges meet at the vertices.

(iii) Yes, a polyhedron can have a square and four triangles as its faces, forming a pyramid with a square base, because all eight edges meet at the vertices.

Question 7

Using Euler's formula, find the values of x, y and z

S.No	Faces	Vertices	Edges
(i)	x	15	20
(ii)	6	y	8
(iii)	14	26	z

Answer

$$(i) F + V - E = 2$$

$$\Rightarrow x + 15 - 20 = 2$$

$$\Rightarrow x - 5 = 2$$

$$\Rightarrow x = 2 + 5$$

$$\Rightarrow x = 7$$

Hence, $x = 7$.

$$(ii) F + V - E = 2$$

$$\Rightarrow 6 + y - 8 = 2$$

$$\Rightarrow y - 2 = 2$$

$$\Rightarrow y = 2 + 2$$

$$\Rightarrow y = 4$$

Hence, $y = 4$.

(iii) $F + V - E = 2$

So,

$$\Rightarrow 14 + 26 - z = 2$$

$$\Rightarrow 40 - z = 2$$

$$\Rightarrow z = 40 - 2$$

$$\Rightarrow z = 38$$

Hence, $z = 38$.

S.No	Faces	Vertices	Edges
(i)	7	15	20
(ii)	6	4	8
(iii)	14	26	38

Question 8

What is the least number of planes that can enclose a solid ? What is the name of the solid ?

Answer

The least number of planes that can enclose a solid is four. The solid formed by these four planes is called a tetrahedron.

Hence, the shape is tetrahedron which has 4 planes.

Question 9

Is a square prism same as a cube ?

Answer

A cube is a three-dimensional solid shape with equal length, width and height.

A square prism has a square base, but its height may not be the same as its base dimensions.

- If the height is equal to the base's dimensions, then the square prism is a cube.
- If the height differs from the base's dimensions, then the square prism is a cuboid.

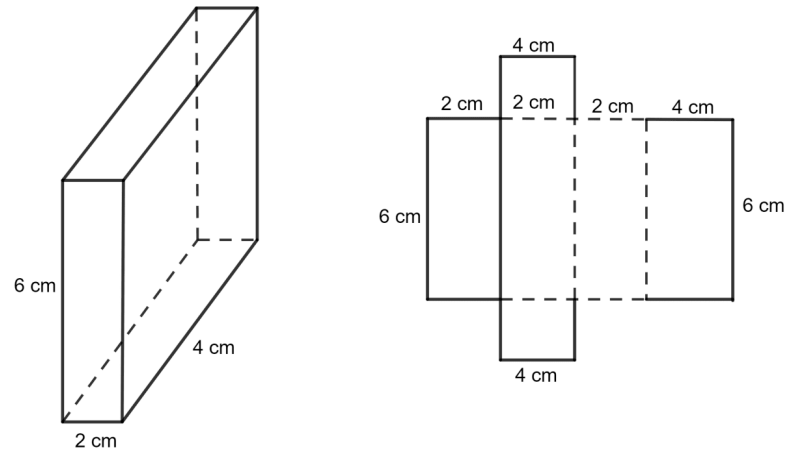
Hence, a square prism can either be a cube or not.

Question 10

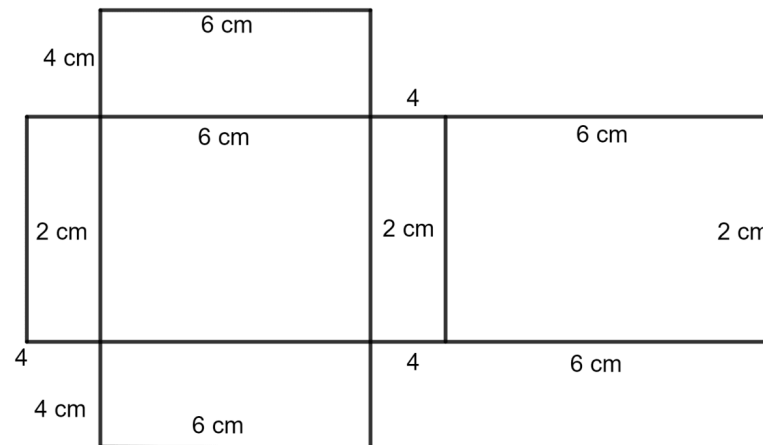
The dimensions of a cuboid are 6 cm x 4 cm x 2 cm. Draw two different nets of it.

Answer Two different nets of the cuboid are shown below:

Net 1

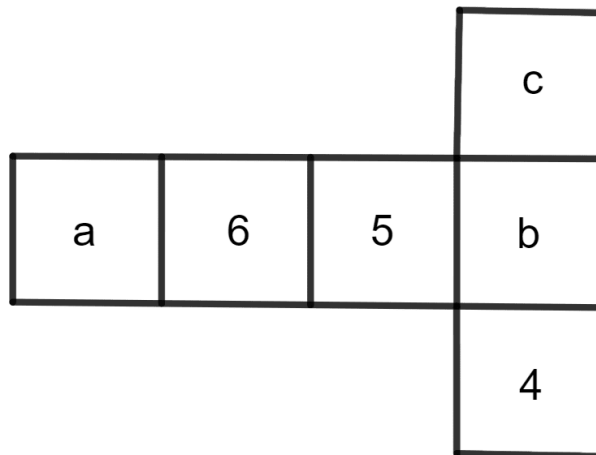


Net 2



Question 11

Dice are cubes where the sum of the numbers on the opposite faces is 7. Find the missing numbers a, b and c.

**Answer**

It is given that the sum of the numbers on the opposite faces is 7.

$$\text{So, } c + 4 = 7$$

$$\Rightarrow c = 7 - 4$$

$$\Rightarrow c = 3$$

$$\text{And, } 5 + a = 7$$

$$\Rightarrow a = 7 - 5$$

$$\Rightarrow a = 2$$

$$\text{And, } 6 + b = 7$$

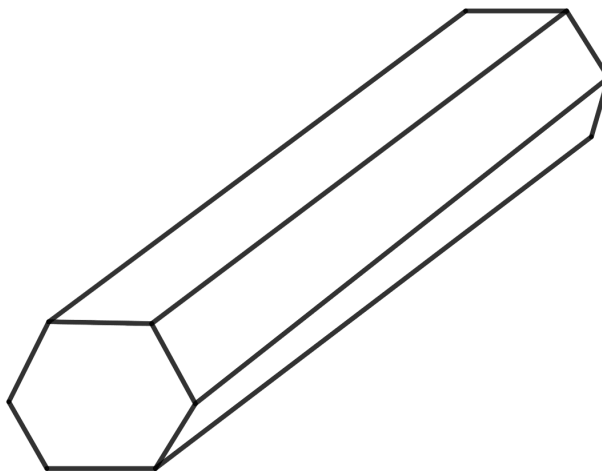
$$\Rightarrow b = 7 - 6$$

$$\Rightarrow b = 1$$

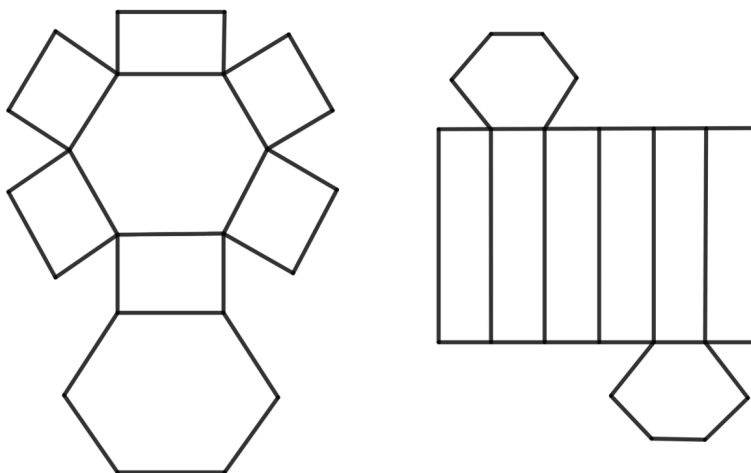
Hence, the values are $a = 2$, $b = 1$ and $c = 3$.

Question 12(i)

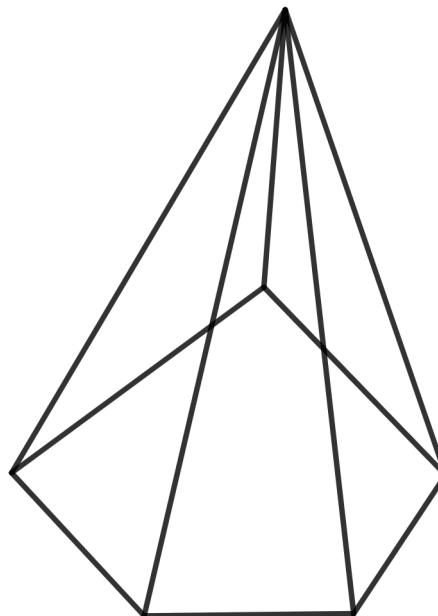
Draw two nets for the following polyhedron :

**Answer**

The two nets are shown below:

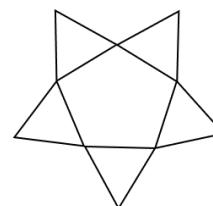
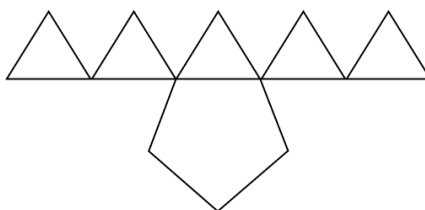
**Question 12(ii)**

Draw two nets for the following polyhedron :



Answer

The two nets are shown below:



Test Yourself

Question 1(i)

If in a polyhedron, number of faces = 20 and number edges = 30; the number of vertices is :

1. 12

2. 6

3. 8

4. 20

Answer**Given:**

$$F = 20, E = 30$$

According to Euler's formula,

$$F + V - E = 2$$

$$\Rightarrow 20 + V - 30 = 2$$

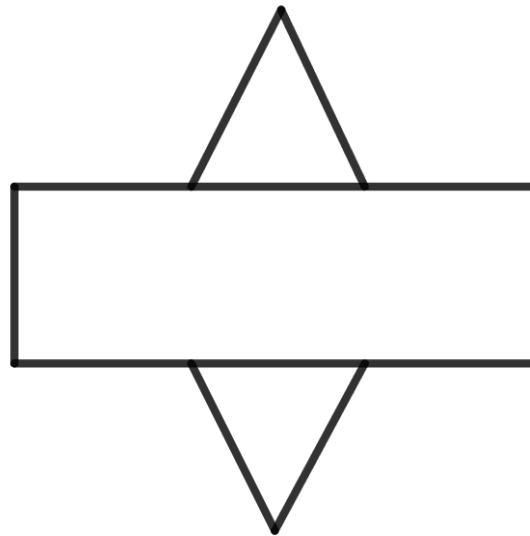
$$\Rightarrow V - 10 = 2$$

$$\Rightarrow V = 2 + 10$$

$$\Rightarrow V = 12$$

Hence, option 1 is the correct option.

Question 1(ii)



is the net of :

1. square pyramid
2. triangular prism
3. hexagonal pyramid

4. none of these

Answer

The given figure is the net of triangular prism.

Hence, option 2 is the correct option.

Question 1(iii)

Joseph is making a pentagonal prism using identical straws. How many straws does he need ?

1. 20

2. 18

3. 15

4. 10

Answer

The number of edges in a pentagonal prism is 15, so Joseph needs 15 identical straws.

Hence, option 3 is the correct option.

Question 1(iv)

In case of a hexagonal pyramid, F = no. of faces and V = no. of vertices, then $F + V$ is:

1. 10

2. 12

3. 14

4. 16

Answer

For a hexagonal pyramid,

$$F = 7, V = 7$$

$$\text{So, } F + V = 7 + 7$$

$$= 14$$

Hence, option 3 is the correct option.

Question 1(v)

The number of faces in a triangular pyramid is :

1. 3

2. 4

3. 5

4. none of these

Answer

Number of faces in a triangular pyramid is 4.

Hence, option 2 is the correct option.

Question 1(vi)

Statement 1: The number of vertices in a pyramid is one more than the number of sides in a polygon.

Statement 2: A polyhedron may have 10 faces, 20 edges and 15 vertices.

Which of the following options is correct?

1. Both the statements are true.
2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.
4. Statement 1 is false, and statement 2 is true.

Answer

The base of a pyramid is a polygon. Let no. of sides in polygon be n , so total number of vertices in a pyramid will be n .

So, the total number of vertices in a pyramid will be $n + 1$.

According to statement 1 :

The number of vertices in a pyramid is one more than the number of sides in a polygon.

So, statement 1 is true.

Given,

A polyhedron may have 10 faces, 20 edges and 15 vertices.

If this is true, then it will satisfy Euler's formula.

Using Euler's formula :

$$F + V - E = 2.$$

Substituting the values in L.H.S., we get

$$\Rightarrow 10 + 15 - 20$$

$$\Rightarrow 25 - 20$$

$$\Rightarrow 5$$

$$\text{R.H.S.} = 2$$

Since, L.H.S. $\neq$ R.H.S.

So, statement 2 is false.

$\therefore$ Statement 1 is true, and statement 2 is false.

Hence, option 3 is the correct option.

Question 1(vii)

Assertion (A) : In a polyhedron, if there are 6 vertices, 12 edges, then the number of faces are 8.

Reason (R) : In a pentagonal pyramid, there are 6 faces, 6 vertices and 10 edges.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.

3. A is true, but R is false.

4. A is false, but R is true.

Answer

In assertion (A),

Given,

$$E = 12 \text{ and } V = 6.$$

Using Euler's formula :

$$\Rightarrow F + V - E = 2.$$

Substituting the values, we get :

$$\Rightarrow F + 6 - 12 = 2$$

$$\Rightarrow F - 6 = 2$$

$$\Rightarrow F = 2 + 6$$

$$\Rightarrow F = 8$$

So, assertion (A) is true.

In reason (R),

Given,

$$F = 6, E = 10, \text{ and } V = 6.$$

Using Euler's formula :

$$\Rightarrow F + V - E = 2.$$

Substituting the values in L.H.S., we get

$$\Rightarrow 6 + 6 - 10$$

$$\Rightarrow 12 - 10$$

$$\Rightarrow 2$$

$$\text{R.H.S.} = 2$$

Since, L.H.S. = R.H.S.

So, reason (R) is true.

$\therefore$ Both A and R are correct, and R is not the correct explanation for A.

Hence, option 2 is the correct option.

Question 1(viii)

Assertion (A) : If a polyhedron has 7 vertices and 10 faces, the number of edges is 19.

Reason (R) : The relationship between faces (F), edges (E) and vertices (V) of a polyhedron is $F + V - E = 2$.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

By Euler's formula, the relationship between faces (F), edges (E) and vertices (V) of a polyhedron is : $F + V - E = 2$.

So, reason (R) is true.

Given,

$$V = 7 \text{ and } F = 10$$

Substituting the values, in Euler's formula we get :

$$\Rightarrow 10 + 7 - E = 2$$

$$\Rightarrow 17 - E = 2$$

$$\Rightarrow E = 17 - 2$$

$$\Rightarrow E = 15$$

So, assertion (A) is false.

$\therefore$ A is false, but R is true.

Hence, option 4 is the correct option.

Question 1(ix)

Assertion (A) : The number of edges in a triangular prism = 9.

Reason (R) : In a triangular prism, the number of vertices = 2 x number of sides
= 6;

The number of faces = 2 + number of sides = 5.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

A triangular prism has 3 edges on each triangular base and 3 edges connecting the bases.

$$\text{Total edges} = 3 + 3 + 3 = 9$$

So, assertion (A) is true.

As we know that

A triangular prism has 5 faces, 9 sides, and 6 vertices.

$$\text{The number of vertices} = 2 \times \text{number of sides} = 2 \times 9 = 18.$$

$$\text{The number of faces} = 2 + \text{number of sides} = 2 + 9 = 11.$$

So, reason (R) is false.

$\therefore$ A is true, but R is false.

Hence, option 3 is the correct option.

Question 1(x)

Assertion (A) : The number of edges in a rectangular pyramid = 8.

Reason (R) : In a triangular prism, the number of vertices is one more than number of sides and the number of faces is one less than number of sides.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are false.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

A rectangular pyramid has 5 faces, 8 edges, and 5 vertices.

So, assertion (A) is true.

A triangular prism has 5 faces, 9 sides, and 6 vertices.

Difference between sides and vertices = $9 - 6 = 3$

Difference between sides and faces = $9 - 5 = 4$

So, reason (R) is false.

$\therefore$ A is true, but R is false.

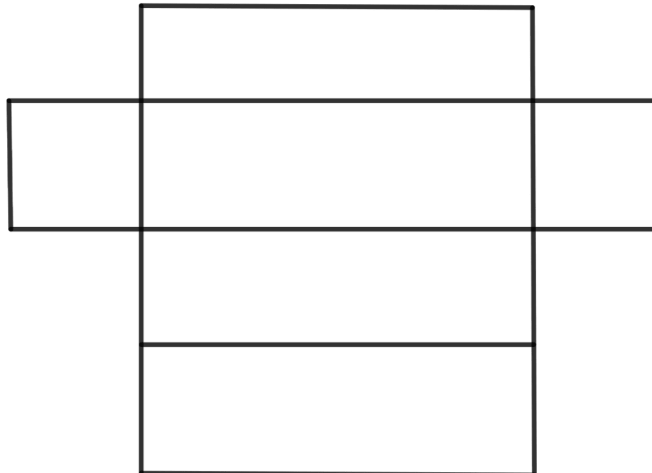
Hence, option 3 is the correct option.

Question 2

Draw the net of a rectangular box.

Answer

Net of a rectangular box is shown below:

**Question 3**

Can a polyhedron have 15 faces, 30 edges and 20 vertices ?

Answer

$$F = 15, V = 20, E = 30$$

According to Euler's formula,

$$F + V - E = 2$$

$$F + V - E = 15 + 20 - 30$$

$$= 35 - 30$$

$$= 5$$

$$\Rightarrow F + V - E \text{ is not equal to } 2.$$

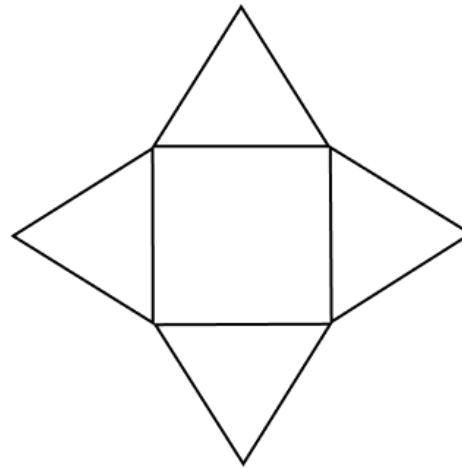
Hence, a polyhedron cannot have 15 faces, 30 edges and 20 vertices.

Question 4

Draw the net of a square pyramid.

Answer

Net of a square pyramid is shown below:

**Question 5**

Write the number of edges in a hexagonal pyramid.

Answer

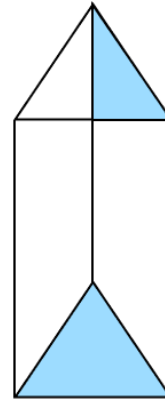
The number of edges in a hexagonal pyramid is 12.

Question 6

Draw the two dimensional representation of a triangular prism.

Answer

Two dimensional representation of a triangular prism is shown below:



Question 7

If a polyhedron has 10 faces and 8 vertices, find the number of edges in it.

Answer

Given:

$$F = 10, V = 8$$

According to Euler's formula,

$$F + V - E = 2$$

$$\Rightarrow 10 + 8 - E = 2$$

$$\Rightarrow 18 - E = 2$$

$$\Rightarrow E = 18 - 2$$

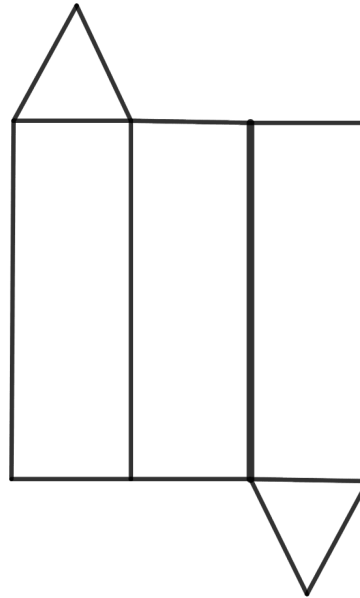
$$\Rightarrow E = 16$$

Hence, the number of edges = 16.

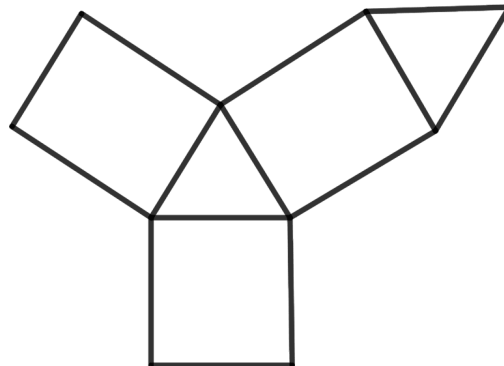
Question 8

Name the polyhedron that can be made by folding each of the following nets :

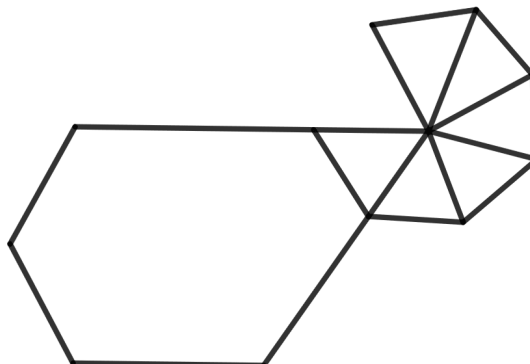
(i)



(ii)



(iii)



Answer

(i) It has 3 rectangles and 2 triangles.

Hence, it is triangular prism.

(ii) It has 3 rectangles and 2 triangles.

Hence, it is triangular prism.

(iii) It has 1 hexagonal base and 6 triangles.

Hence, it is hexagonal pyramid.

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